Thieme

Effects of Intramuscular Midazolam and Lorazepam on Acute Agitation in Non-Elderly Subjects – A Systematic Review

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Key words
agitation, antipsychotics, lorazepam, midazolam

received 04.11.2016
revised 05.01.2017
accepted 09.01.2017

Bibliography
DOI http://dx.doi.org/10.1055/s-0043-100766
Published online: 2017 | Pharmacopsychiatry
© Georg Thieme Verlag KG Stuttgart · New York
ISSN 0176-3679

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ABSTRACT

Introduction Benzodiazepines are commonly used for the treatment of acute agitation in a psychiatric setting.
Methods We searched MEDLINE, EMBASE, PsycINFO, and the Cochrane Central Register of Controlled Trials (CENTRAL) for relevant publications. Randomized trials evaluating intramuscular (IM) midazolam or lorazepam given as monotherapy or as add-on treatment, with more than 10 patients aged 18–65 years, conducted in a psychiatric setting, and published between January 1, 1980, and February 3, 2016, were included.
Results 16 studies from a search result of 5 516 studies were included. In total, 577 patients were treated with lorazepam IM 2–4 mg, and 329 patients were treated with midazolam IM 5–15 mg.
Discussion It is unclear whether lorazepam IM or midazolam IM is as efficacious as an antipsychotic IM. It is a bit more certain that the combination of benzodiazepines IM and a low dose antipsychotic IM is more efficacious than the benzodiazepine and the antipsychotic alone. However, there is no doubt that benzodiazepines are less likely to be associated with treatment emergent side effects, as compared to antipsychotics.

Introduction

According to the glossary of definitions as part of the Schedules for Clinical Assessment in Neuropsychiatry (SCAN) [1], agitation is excessive motoric activity with a background of marked anxiety, distinguishable from gross excitement in which the subject is much wilder and often aggressive [1]. In the trial literature, most often both of these behavioral characteristics are referred to as severe agitation [2]. Usually, agitation develops as part of a psychiatric disorder such as mania or schizophrenia, but it may also occur on its own. The severely agitated person may be extremely aroused, tense, hostile, or confused. Due to the action of the agitated person, interventions are often needed and can be verbal or nonverbal de-escalation, psychotropic drugs, unlocked or locked seclusion, restraints, or a combination of these, depending on the setting [2]. The most commonly used psychotropic drug classes for the treatment of agitation are antipsychotics and benzodiazepines, administered orally or intramuscularly [3]. For the most severely agitated patients, the use of intramuscular formulations is often needed to achieve a faster onset of effect and due to lack of patient collaboration regarding medication intake. When applying antipsychotics acutely, in particular typical antipsychotics, extrapyramidal side effects (EPS) in the form of acute dystonia or Parkinsonism and cardiac side effects (e. g., QT prolongation and orthostatic hypotension) may develop. The acute use of benzodiazepines may lead to cognitive impairment, decreased motor coordination, dizziness [4], and in rare cases (during use of high doses) respiratory depression [5–7]. Spontaneous reporting of side effects, though, could be low [8]. The beneficial effects of antipsychotics on agitation are considered to be mediated through their dopaminergic action, resulting in a specific dampening and, for some of the antipsychotics, through an additional antihistaminergic action resulting in sedation. The benzodiazepines act through a positive allosteric modulation on the GABA receptor, resulting in anxiolytic and sedative effects.

In accordance with what is outlined above, antipsychotics and benzodiazepines or their combination are the treatments most often evaluated in clinical trials, and for that reason, most reviews of the efficacy and tolerability of various pharmacological interventions for agitation have focused on these treatments [2, 3, 9]. Several benzodiazepines have been tested for the treatment of agitation, but only midazolam and lorazepam are readily absorbed after

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intramuscular administration without carrying the risk of causing significant tissue damage [10].

In this systematic review, we aimed at investigating the efficacy and tolerability of intramuscular lorazepam and midazolam (used either alone or in combination with an antipsychotic) for the treatment of severe agitation in psychiatric settings.

Methods

Selection criteria

Randomized trials evaluating intramuscular (IM) midazolam or lorazepam given as monotherapy or as add-on treatment, with more than 10 patients aged 18-65 years, conducted in a psychiatric setting, and published between January 1, 1980, and February 3, 2016, were included. Trials published only as conference abstracts were also considered. Only studies statistically comparing treatment groups were included.

Information sources (search strategy)

We searched for relevant publications in MEDLINE, EMBASE, PsycINFO, and the Cochrane Central Register of Controlled Trials (CENTRAL). In addition, the Cochrane Database of Systematic Reviews (CDSR), the Database of Abstracts of Reviews of Effects (DARE), and the Cochrane Health Technology Assessment (HTA) database were searched for relevant secondary publications. A hand search of primary and secondary publication reference lists was conducted to identify further relevant studies.

Search string

((((“Midazolam”[Mesh]) OR Midazolam) OR (“Ro 21-3981” OR “Ro 21 3981” OR “Ro 213981” OR Dormicum OR versed))) AND ((“Psychomotor Agitation”[Mesh]) OR (Akathisia OR Restlessness OR Psychomotor AND (Hyperactivity OR agitation OR excitement)))

(((“Lorazepam”[Mesh]) OR (((((((((((((((lorazepam) OR Ativan) OR “Orfidal Wyeth”) OR Donix) OR Duralozam) OR Idalprem) OR Laubeel) OR lorazep) OR “Novo Lorazem”) OR “Nu-Loraz”) OR Nu-Loraz) OR Sedicepan) OR Sinestron) OR Somagerol) OR Temesta) OR Tolid) OR (“WY-4036” OR “WY 4036” OR WY4036)))) AND ((“Psychomotor Agitation”[Mesh]) OR (Akathisia OR Restlessness OR Psychomotor AND (Hyperactivity OR agitation OR excitement)))

The first search was performed on June 3, 2013, and updated last on February 2, 2016. The search was conducted in collaboration with Connie Skubbeltrang at Medical Library, Aalborg University Hospital, Denmark.

Study selection process

Author SJK conducted the initial literature screening. Authors SJK and REN reviewed and selected studies manually, first according to title and abstract and later to full-text for inclusion in the systematic review independently of each other. Discrepancies between the 2 were discussed openly, and agreement on inclusion was then met.

Outcome measures

No consensus has been reached regarding outcome evaluation in studies of acute agitation. As a result, several outcome measures are commonly used, as demonstrated in the included studies. To compare treatment groups within each study, we defined superiority as invention A being statistical superior to invention B, with a significance threshold of $P < 0.05$. Similarly, we defined inferiority as intervention A being statistical inferior to invention B, with a significance threshold of $P < 0.05$. Lastly, we defined any differences not reaching statistical significance as no difference between intervention A and intervention B. Based on such individual study evaluations, the results were grouped across studies.

Results

16 studies from an initial search result of 5 516 studies were included in the systematic review (numbered 1–16 in ▶ **Table 1**). The study selection can be seen from a PRISMA flowchart (▶ **Fig. 1**).

Effect

Outcome regarding the effect and the characteristics of the trial and the study population for each of the selected trials are displayed in ▶ **Table 1**.

Lorazepam IM

A total of 577 patients were treated with lorazepam IM 2–4 mg, and of these 300 received the drug as monotherapy. One study compared lorazepam IM monotherapy to placebo and found superiority (study 3). In the 6 comparisons of lorazepam IM monotherapy to an antipsychotic IM monotherapy (studies 1, 2, 3, 7, 8, and 16), 2 found comparable efficacy (studies 1 and 2), 3 found inferiority (studies 3, 7, and 8), and 1 found superiority (study 16). Additionally, lorazepam IM was inferior to a combination of haloperidol IM and the antihistaminergic drug promethazine IM (study 4). In the 1 study comparing lorazapam IM to midazolam IM, no difference was demonstrated (study 16). Lorazapam IM in combination with an antipsychotic was evaluated in 5 studies (studies 5–10). In 2 of the studies, the combination was compared to the antipsychotic IM alone (studies 7 and 8), and in 3 of the studies, the combination was compared to lorazepam IM alone (studies 7–9); in both cases, the combination was superior to monotherapy. In the remaining 2 evaluations of a combined treatment, lorazepam IM plus oral haloperidol was superior to lorazepam IM plus oral risperidone (study 5), and lorazepam IM plus haloperidol IM was superior to olanzapine IM (study 6).

Midazolam IM

A total of 329 patients were treated with midazolam IM 5–15 mg, and of these 275 received the drug as monotherapy. No study compared midazolam IM monotherapy to placebo. Among the 4 comparisons of midazolam IM monotherapy to an antipsychotic IM monotherapy (studies 11, 13, 15, and 16), 1 found comparable efficacy (study 15) and 3 found superiority (studies 11, 13, and 16). Additionally, midazolam IM was superior to a combination of haloperidol IM and promethazine IM (study 12). Midazolam IM in combination with an antipsychotic IM was evaluated in 2 studies (studies 14 and 15). In 1 of the studies, the combination was superior to the antipsychotic combined with promethazine IM (study 14), and in the other, the combination was comparable to both drugs given as monotherapy in double doses (study 15).

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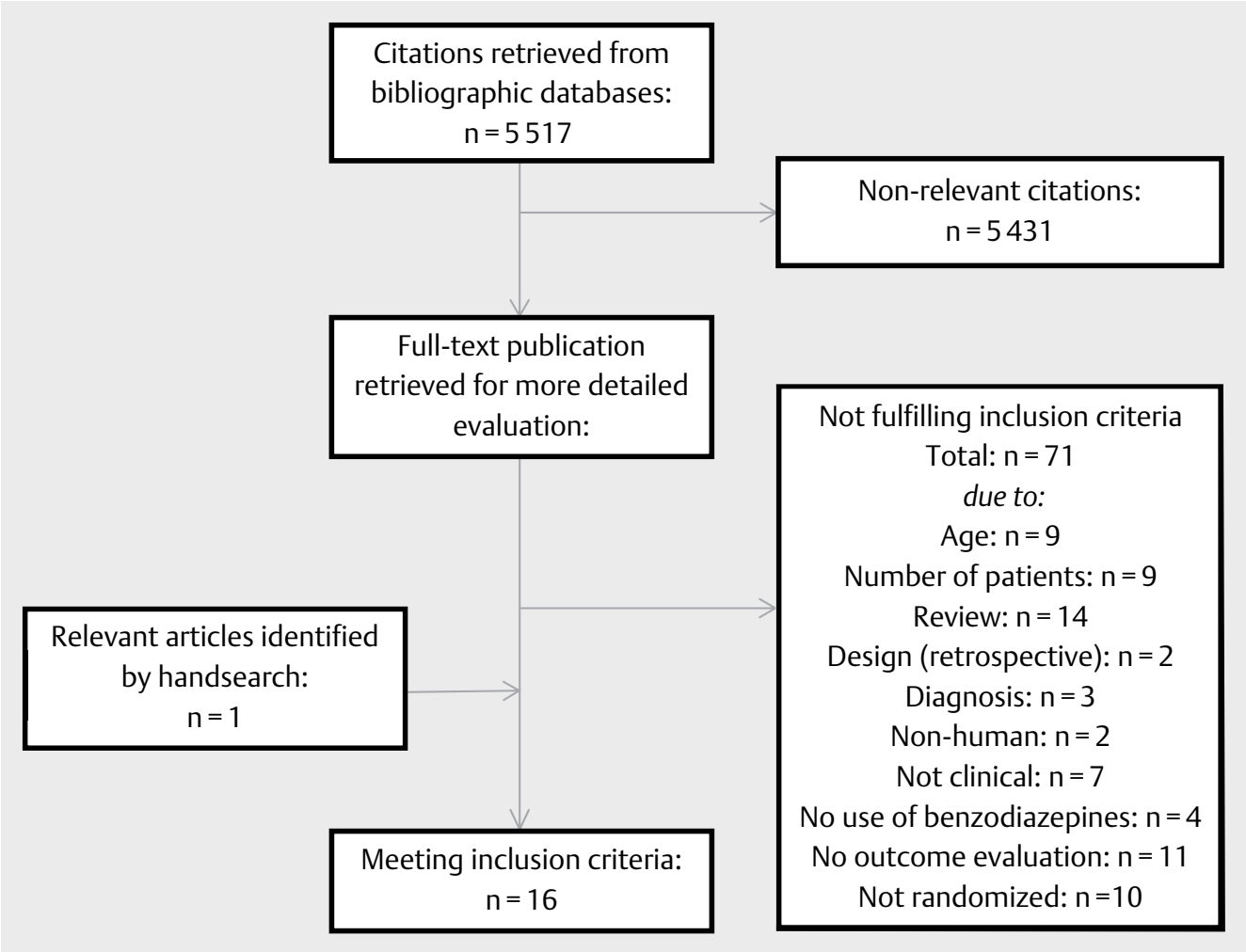

▶ **Fig. 1** Flowchart: study selection.

Effect and study characteristics

In 9 studies, patients with either schizophrenia or mania, or a mix, were enrolled, whereas in the remaining studies, severely ill patients with psychosis and disruptive behavior, but with no further specification of diagnoses, were enrolled. Across all studies, no clear separation between studies with and studies without well-defined diagnostic entry criteria in terms of relative effect of the benzodiazepine was seen. This also applied when focusing on the studies comparing the benzodiazepine in monotherapy with an antipsychotic in monotherapy. 9 of the studies were double-blinded and 5 were single-blinded, but again, no pattern regarding blinding and outcome could be observed. Study duration of agitation evaluation varied considerably with 4 studies lasting 24 h, 3 studies lasting 4–6 h, and the remaining lasting 2 h or less, but study results were also comparable regardless of study duration. Likewise, the sample sizes ranged widely, seemingly without influencing the results. The comparisons indicating a difference between the comparators were seen along the whole range. Among the 4 trials comparing lorazepam monotherapy with haloperidol monotherapy, the 2 positive trials (studies 7 and 8) had sample sizes of 44 and 64, and the 2 negative trials (studies 1 and 2) had sample sizes of 37 and 60. Finally, the sex ratio (about twice as many males as females were included in 10 of the studies) seemed to have no impact on the results. The mean age was relatively low across all trials (43 years or below).

Adverse effects

Lorazepam

Only 5 of the lorazepam studies provided information on adverse effects (studies 1, 2, 6, 8, and 16); the other studies had no description of adverse effects. Adverse effects reported in more than 5 % of the patients receiving lorazepam monotherapy were extreme sedation (18 % in study 2), ataxia, dizziness, dry mouth, and speech disorder. Similar side effects, albeit a bit more frequent, were observed when lorazepam was combined with an antipsychotic. No serious adverse events were observed in patients treated with lorazepam.

Midazolam

5 of the midazolam studies provided information on adverse effects (studies 12–16). In patients receiving midazolam monotherapy, drop in oxygen saturation was seen in 24 % of the patients, and sedation was seen in 62 % of the patients (study 15). Respiratory failure was observed in 1 patient (study 12).

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▶ **Table 1** Study characteristics

Study #	Study	Mean age (years)	Study duration	Diagnosis	N (n male/n female)	Intervention and effect	Outcome measure	Study design
Lorazepam monotherapy								
1	Salzman et al. 1991 [20]	30.5	24 h	Psychotic disruptive behavioural patients	60 (unknown sex distribution)	LOR (IM) 2 mg (n = 30) ≈ HAL (IM) 5 mg (n = 30)	Overt Aggression Scale	Randomized, double-blinded
2	Foster et al. 1997 [21]	43.0	4 h	Schizophrenia	37 (26 males/11 females)	LOR (IM) 2 mg (n = 17) ≈ HAL (IM) 5 mg (n = 20)	Brief Psychiatric Rating Scale, Clinical Global Impression	Randomized, double-blinded
3	Meehan et al. 2001 [22]	39.0	24 h	Bipolar mania	201 (106 males/95 females)	LOR (IM) 2 mg (n = 51) < OLA (IM) 10 mg (n = 99) > placebo (n = 51)	Positive and Negative Syndrome Scale – Excited Component	Randomized, double-blinded, placebo-controlled
4	Alexander et al. 2004 [23]	32.2	240 min	Severe mental illness	200 (119 males/81 females)	LOR (IM) 4 mg (n = 100) < HAL (IM) 10 mg + PRO (IM) 20–50 mg (n = 100)	How many patients were patients tranquil or asleep	Randomized, single-blinded
Lorazepam combination therapy								
5	Currier et al. 2004 [24]	39.2	120 min	Schizophrenia	162 (105 males/57 females)	HAL (oral) 5 mg + LOR (IM) 2 mg (n = 79) > RIS (oral) 2 mg + LOR (oral) 2 mg (n = 83)	Positive and Negative Syndrome Scale	Randomized, rater-blinded
6	Huang et al. 2015 [25]	41.3	24 h	Schizophrenia, bipolar mania	67 (41 males/26 females)	HAL (IM) 5 mg + LOR (IM) 2 mg (n = 30) > OLA (IM) 10 mg (n = 37)	Positive and Negative Syndrome Scale – Excited Component, Agitation-Calmness Evaluation Scale	Randomized, open-label
Lorazepam monotherapy and combination therapy								
7	Garza-Trevino et al. 1989 [26]	31.0	60 min	Schizophrenia, mania	68 (41 males/27 females)	HAL (IM) 5 mg + LOR (IM) 4 mg (n = 24) > HAL (IM) 5 mg (n = 21) > LOR (IM) 4 mg (n = 23)	Agitation scale from 0–100	Randomized, open-label
8	Battaglia et al. 1997 [27]	33.9	24 h	Schizophrenia, mania	98 (73 males/25 females)	LOR (IM) 2 mg + HAL (IM) 5 mg (n = 32) > HAL (IM) 5 mg (n = 35) > LOR (IM) 2 mg (n = 31)	Agitated Behavior Scale, Modified Brief Psychiatric Rating Scale, Clinical Global Impression	Randomized, double-blinded
9	Bieniek et al. 1997 [28]	35.0	60 min	Schizophrenia, bipolar mania	20 (13 males/7 females)	LOR (IM) 2 mg (n = 11) < HAL (IM) 5 mg + LOR (IM) 2 mg (n = 9)	Overt Aggression Scale	Randomized, double-blinded
10	Veser et al. 2006 [29]	35.8	90 min	Severe mental illness	30 (23 males/7 females)	LOR (IM) 2 mg + placebo (n = 10) ≈ RIS (oral) 2 mg + LOR (IM) 2 mg (n = 10) ≈ HAL (oral) 5 mg + LOR (IM) 2 mg (n = 10)	Positive and Negative Syndrome Scale	Randomized, double-blinded.
Midazolam monotherapy								
11	Wyant et al. 1990 [30]	35.6	2 h	Schizophrenia	15 (15 males/0 females)	MID (IM) 5 mg (n = 5) > HAL (IM) 5 mg (n = 5) > SA (IM) 250 mg (n = 5)	Clinical Global Rating Scale	Randomized, patient single-blinded
12	Huf et al. 2003 [31]	38.0	120 min	Psychotic aggrasive behavioural patients	301 (146 males/155 females)	MID (IM) 7.5–15 mg (n = 151) > HAL (IM) 5–10 mg + PRO (IM) 25–50 mg (n = 150)	Time to tranquilization or sleep	Randomized, single-blinded
13	Martel et al. 2005 [32]	36.9	120 min	Psychotic aggressive patients	144 (98 males/46 females)	MID (IM) 5 mg (n = 48) > DRO (IM) 5 mg (n = 50) > ZIP (IM) 20 mg (n = 46)	Altered Mental Status Scale	Randomized, double-blinded
Midazolam combination therapy								

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▲ **Table 1** Continued.

Study #	Study	Mean age (years)	Study duration	Diagnosis	N (n male/n female)	Intervention and effect	Outcome measure	Study design
14	Mantovani et al. 2013 [33]	32.9	90 min	Schizophrenia, bipolar mania	100 (47 males/53 females)	HAL (IM) 2.5 mg + MID (IM) 7.5 mg (n = 25) > HAL (IM) 2.5 mg + PRO (IM) 25 mg (n = 27) > ZIP (IM) 10 mg (n = 23) > OLA 10 mg (n = 25)	Calming Agitation-Evaluation Scale	Randomized, rater-blinded
Midazolam mono- and combination therapy								
15	Isbister et al. 2010 [34]	35.0	6 h	Psychotic disruptive patients	91 (45 males/46 females)	DRO (IM) 5 mg + MID (IM) 5 mg (n = 29) ≈ MID (IM) 10 mg (n = 29) ≈ DRO (IM) 10 mg (n = 33)	Duration of violent and acute behavioral disturbance	Randomized, double-blinded
Lorazapam and Midazolam monotherapy								
16	Nobay et al. 2004 [35]	39.5	4 h	Psychotic disruptive patients	111 (unknown sex distribution)	LOR (IM) 2 mg (n = 27) ≈ MID (IM) 5 mg (n = 42) > HAL (IM) 5 mg (n = 42)	Time to sedation	Randomized, double-blinded

LOR (lorazepam), MID (midazolam), HAL (haloperidol), RIS (risperidone), PRO (promethazine), DRO (droperidol), ZIP (ziprasidone), SA (sodium amytal), OLA (olanzapine). > (statistically significant superior; $P < 0.05$), < (statistically significant inferior; $P < 0.05$), ≈ (no statistically significant difference)

Antipsychotics

In patients receiving an antipsychotic in monotherapy (primarily haloperidol or droperidol), akathisia or dystonia were observed relatively frequently. In 1 study, extrapyramidal side effects were observed in 61 % of the patients (study 14). In another study, 2 out of 42 patients treated with haloperidol developed hypotensive and apnoeic adverse effects (study 16). In the patients receiving a combination of an antipsychotic and either lorazepam or midazolam, the frequency of extrapyramidal side effects seemed to be lower than in the patients receiving the antipsychotic alone.

Discussion

In this systematic review, we included 16 randomized trials evaluating lorazepam IM or midazolam IM given either as monotherapy or in combination with an antipsychotic. In total, 575 patients were randomized to monotherapy with either lorazepam or midazolam (300 treated with lorazepam and 275 treated with midazolam), and 331 patients were randomized to either lorazepam or midazolam in combination with an antipsychotic (277 treated with lorazepam in combination with an antipsychotic and 54 treated with midazolam in combination with an antipsychotic). With a few exceptions, the chosen antipsychotics were either haloperidol or droperidol (i. e., 2 typical antipsychotics carrying only a low risk of causing hypotension). The studies were highly heterogeneous, in particular regarding treatment arms and methods for outcome evaluations, with some studies not even applying a quantitative measure, making meta-analysis impossible. In fact, the differences in comparisons even make an overall qualitative evaluation difficult. Lorazepam IM was superior to placebo, and based on the monotherapy comparisons of lorazepam IM with an antipsychotic IM, lorazepam seems less (2 out of 4 studies) or equally effective to the antipsychotics. Furthermore, the combination of lorazepam IM and haloperidol IM was superior to haloperidol or lorazepam alone. In 2 trials (studies 5 and 6), the role of lorazepam in combination with haloperidol is impossible to interpret since haloperidol is dosed higher regarding dopamine blockade than the antipsychotic comparator. Midazolam IM has not been tested against placebo, but in monotherapy comparisons with an antipsychotic IM, midazolam was superior (3 out of 4 studies) or equal to the antipsychotic. In addition, midazolam IM was superior to a combination of droperidol IM plus promethazine IM. The 2 studies in which midazolam IM was combined with an antipsychotic suggest that adding midazolam IM to an antipsychotic IM is superior to adding promethazine IM (study 14) and that the effect of midazolam IM and droperidol IM is additive (study 15). Regarding the effect of lorazepam IM vs. the effect of midazolam IM, 1 comparison could not detect any difference (study 16). However, an indirect comparison based on the comparisons of each of the drugs in monotherapy with an antipsychotic in monotherapy might suggest that midazolam could be more effective than lorazepam. As the pharmacodynamics effects of benzodiazepines are considered to be similar, differences in effect are most likely explained by dosing, since benzodiazepines are known to differ only regarding pharmacokinetic properties [11].

The characteristics including the quality of the selected trials varied considerably. However, no relationship between specific

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characteristics and trial results could be identified. The sample sizes were around 100 or above in 5 studies and below 25 in 4 studies. Type 2 errors could always have occurred, but it is noteworthy that differences in outcome were seen across the entire range of sample sizes. On the other hand, results may be inflated due to publication bias and due to multiple testing within some of the trials.

When generalizing the findings, the relative low age (mean age 43 years or below) should be emphasized. Obviously, it reflects that trials enrolling patients above 65 years of age were not selected, since the treatment of agitation in elderly patients follows other principles [12]. Despite the fact that the enrolled study populations were not uniform regarding formal diagnosis, they all consisted of severely agitated patients, and since there were no differences in results across the studies, the generalizability is most likely not affected by diagnosis per se. Furthermore, even though the most severely agitated and psychotic patients most likely were not among the recruited patients, the results might be applicable to such patients as well.

Based on the proportion of the studies, which provided data on tolerability, lorazepam and midazolam were relatively well tolerated, even though various cognitive and motor side effects occurred in a substantial fraction of the patients. However, it should be noted that 1 patient treated with midazolam experienced respiratory failure and that midazolam reduced oxygen saturation in a relatively high proportion of patients. No information on oxygen saturation was provided in the lorazepam studies, but generally, this potential adverse event should lead to caution regarding using doses of the benzodiazepines higher than those used in the trials. Also as expected, the benzodiazepines were better tolerated than the antipsychotics due to the absence of extrapyramidal side effects. Our finding of a more favorable side effect profile of benzodiazepines as compared to antipsychotics was similar to the result of another recent review by Kishi et al. [13]. A number of studies have attributed a variety of side effects to long-term treatment with benzodiazepines [14–19], but obviously, the burden of side effects are reduced when the drugs are used only acutely.

In summary, based on our review, it is unclear whether lorazepam IM or midazolam IM is as efficacious as an antipsychotic IM. It is a bit more certain that the combination of 1 of these benzodiazepines IM and a low dose typical antipsychotic IM is more efficacious than the benzodiazepine and the antipsychotic given as monotherapy. However, there is no doubt that the benzodiazepines are less likely to be associated with treatment emergent side effects, as compared to antipsychotic drugs. Translating this into clinical practice, a benzodiazepine IM should be chosen ahead of an antipsychotic IM, at least ahead of a typical antipsychotic. Addition of benzodiazepine treatment in patients already treated with antipsychotic drugs, who experience acute agitation, is also supported by the results in the current review.

In conclusion, we find both lorazepam IM (dose range 2–4 mg) and midazolam IM (dose range 5–15 mg) to be suitable additions to the armamentarium of psychopharmacological treatment of non-elderly acutely agitated patients

Acknowledgements

The authors would like to acknowledge Conni Skrubbeltrang at Medical Library, Aalborg University Hospital for her help in constructing and executing the search.

Funding

The affiliated research institution has funded the study. The corresponding author had full access to all the data in the study and took the final decision to submit for publication.

Conflicts of Interest

SJ Kousgaard has nothing to declare. RW Licht has received consulting fees and honoraria for lecturing within the last 3 years from Astra-Zeneca, Bristol-Myers Squibb, Janssen Cilag, Lundbeck, Otsuka, Servier, and Sunovion. RE Nielsen has received research grants from H. Lundbeck for clinical trials, received speaking fees from Bristol-Myers Squibb, Astra Zeneca, Janssen & Cilag, Lundbeck, Servier, Otsuka Pharmaceuticals, and Eli Lilly, and has acted as advisor to Astra Zeneca, Eli Lilly, Lundbeck, Otsuka Pharmaceuticals, Takeda, and Medivir.

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